OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 30 OF 55

Extended Abstract: AI-Based IoT Device Anomaly Detection for Smart Home Security

Domain: IoT Telematics & Edge Anomaly Detection | **Module:** project_30_iot_anomaly

OFFICIAL ABSTRACT

AI-Based IoT Device Anomaly Detection for Smart Home Security is a comprehensive and crucial project aimed at addressing the mounting cybersecurity threats present in modern connected residential environments. As the proliferation of Internet of Things (IoT) devices accelerates, households increasingly rely on interconnected systems that handle sensitive personal data, making them prime targets for malicious actors. Traditional security methods often rely on static rules and manual monitoring, which can be time-consuming, resource-intensive, and fundamentally inefficient against rapidly evolving, sophisticated cyber-attacks and zero-day intrusions. To address these persistent challenges and limitations, this study proposes an advanced, multi-layered approach combining cutting-edge deep learning techniques with intelligent machine learning algorithms for efficient, automated, and real-time IoT anomaly detection. The system utilizes specialized deep learning architectures to continuously capture and extract key behavioral features from continuous network traffic and device telemetry, such as complex packet patterns, fluctuating connection frequencies, irregular payload sizes, and subtle operational deviations. Once these underlying structural patterns are mapped, the extracted features are fed into classification and clustering algorithms designed to enhance decision-making, optimize computational overhead, improve detection accuracy, and drastically reduce false alarm and misclassification rates. By profiling what constitutes normal, routine behavior for each distinct device within the home network, the model establishes a dynamic baseline that adapts to changing household usage patterns over time. AI-Based IoT Device Anomaly Detection for Smart Home Security is a comprehensive and crucial project aimed at addressing the mounting cybersecurity threats present in modern connected residential environments. As the proliferation of Internet of Things (IoT) devices accelerates, households increasingly rely on interconnected systems that handle sensitive personal data, making them prime targets for malicious actors. Traditional security methods often rely on static rules and

manual monitoring, which can be time-consuming, resource-intensive, and fundamentally inefficient against rapidly evolving, sophisticated cyber-attacks and zero-day intrusions. To address these persistent challenges and limitations, this study proposes an advanced, multi-layered approach combining cutting-edge deep learning techniques with intelligent machine learning algorithms for efficient, automated, and real-time IoT anomaly detection.

📌 Project Summary & System Specifications

Project Title	Extended Abstract: AI-Based IoT Device Anomaly Detection for Smart Home Security
Specialization / Category	IoT Telematics & Edge Anomaly Detection
Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5030`
Local Implementation	`project_30_iot_anomaly`